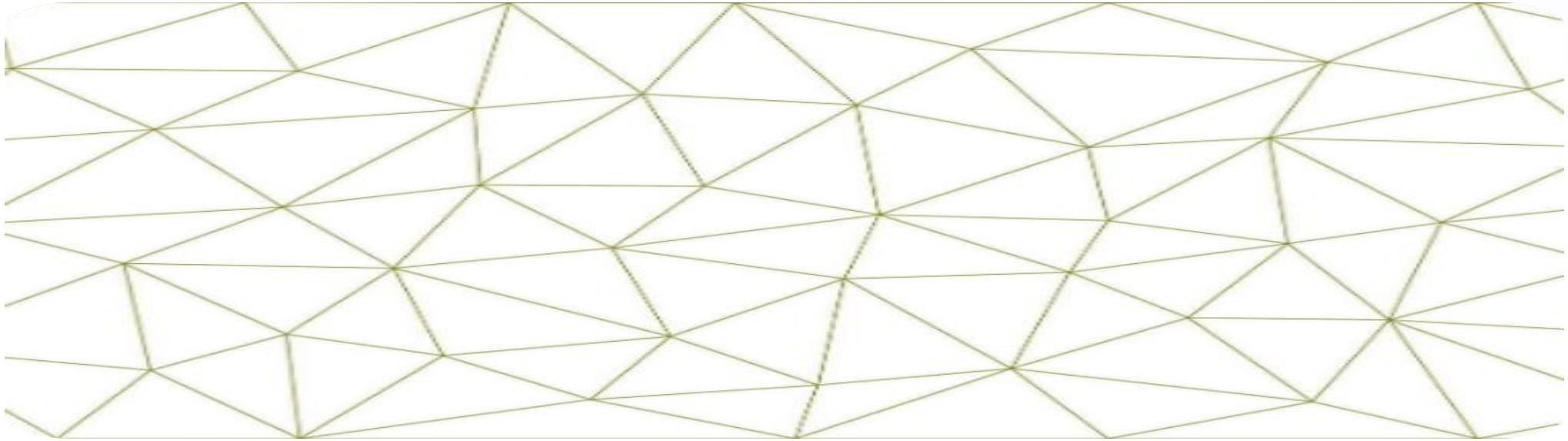
# Loan Approval Prediction System



A Project by Ashish Giri | IBM SkillsBuild

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# Problem Statement

### Operational Lag

Traditional manual underwriting is time-consuming, creating bottlenecks that degrade the borrower experience and increase institutional costs.

### Inherent Bias

Subjective human judgment leads to inconsistent approvals and potential demographic biases, affecting financial fairness.

**Requirement: A data-driven, real-time automated system to mitigate risk and ensure unbiased scalability.**

# OBJECTIVES

- **Data Optimization:** Process Clean and engineer historical financial data to identify key credit risk indicator.
- **Predictive Modeling:** Train and compare algorithms (Logistic Regression, Random Forest , XGBoost) to an accurately predict loan defaults.
- **Handling Data Imbalance:** Train Apply algorithms weighting techniques to ensure the system successfully detects high risk applicants.
- **System Deployment:** Develop an interactive web dashboard to provide real-time, automated loan approval decisions.

## Real-Time UI

- Develop an interactive web-based application that enables user to enter the financial information and receive instant lone eligibility predictions.
- Implement dynamic chart and visual risk assessments to clearly display probability scores and key contributing factors.
- Ensures seamless data collection through the user interface to continuously feed and optimize model training pipelines for improved predictive accuracy .

# DATASET DESCRIPTION

The system leverages a comprehensive financial dataset with 20,000 records across 22 attributes.

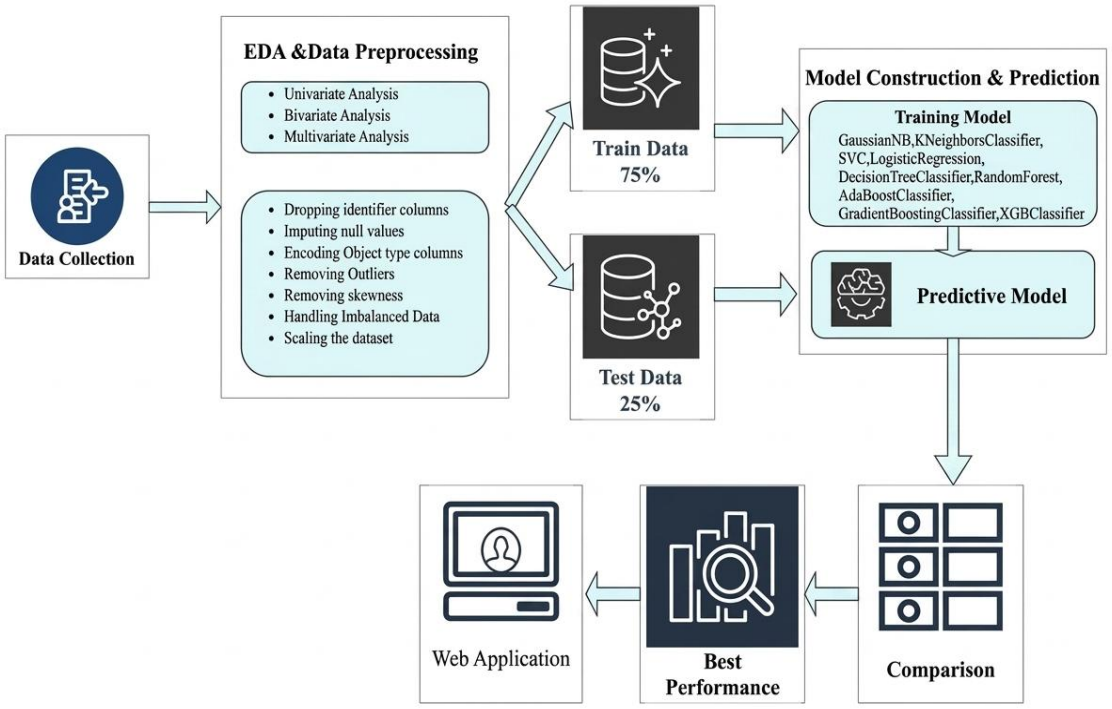
| Feature Category | Examples | Value Range / Type |
|---|---|---|
| Demographic | Age, Gender, Education | 21-70 Years, Discrete |
| Financial | Annual Income, Debt-Ratio | Up to $400k, Continuous |
| Credit History | Credit Score, Delinquency | 0-750, Historical Log |
| Target Variable | Loan Paid Back | Binary (0: Default, 1: Paid) |

# METHODOLOGY

## End-to-End Pipeline

An 8-phase lifecycle was implemented, transforming raw data collection into a serialized predictive model integrated with a web API.

1. Data Collection
2. EDA & Data Preprocessing
3. Train and Test Data
4. Model Construction & Prediction
5. Evaluation, Deployment & Prediction



# DATA PREPROCESSING

### Statistical Imputation

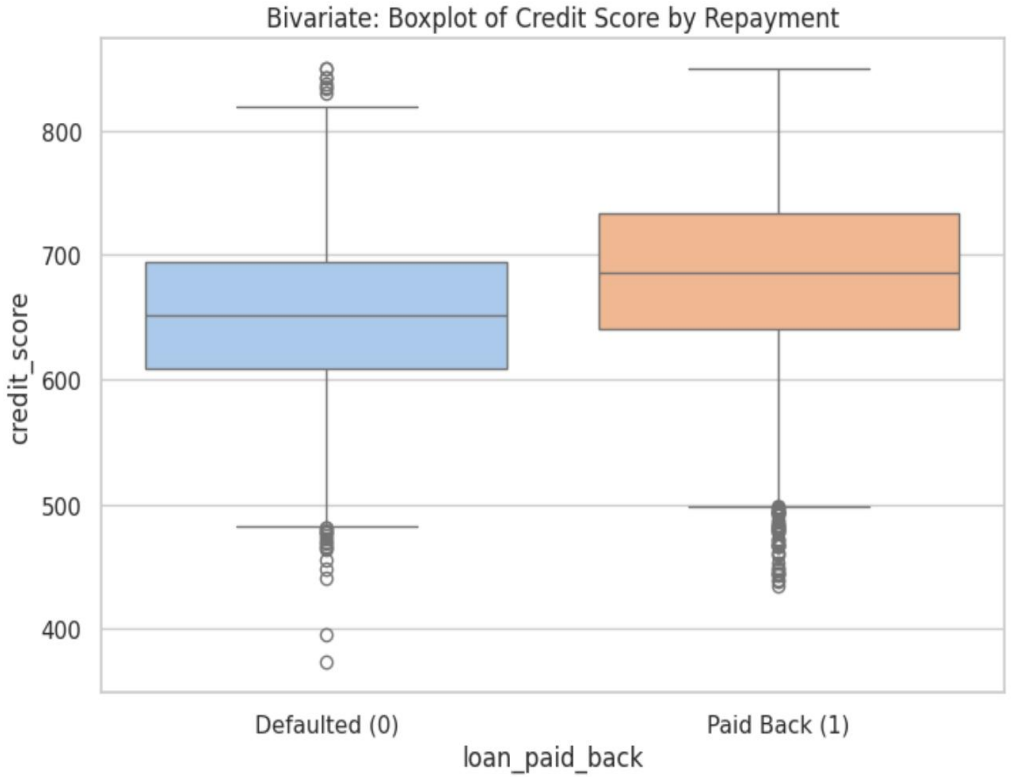
Missing values were handled using median imputation for numerical features and mode imputation for categorical features.

### Feature Scaling

Applied Label Encoding and Standard Scaler to prepare data for effective training and prediction. Outliers and inconsistent records were handled to improve data quality.

**Result:** A clean, consistent and machine-learning-ready dataset was obtained, enabling accurate training and evaluation of loan approval predication models.

# EXPLORATORY DATA ANALYSIS




## Key Indicators

EDA revealed distinct separations in creditworthiness indicators:

- Credit Score: Defaults averaged 651 vs. 686 for successful repayment.
- Outliers: Income showed heavy right skewness, requiring capping via the IQR method.
- Debt-Ratio: Strongest negative correlation (r = -0.223) with success.

# MODELS USED



## Logistic Regression

Our statistical baseline. Excellent interpretability, establishing a linear decision boundary via sigmoid mapping.



## Random Forest

Bagging ensemble of 100 decision trees. Reduced variance by averaging uncorrelated predictions from bootstrap samples.



## XGBoost

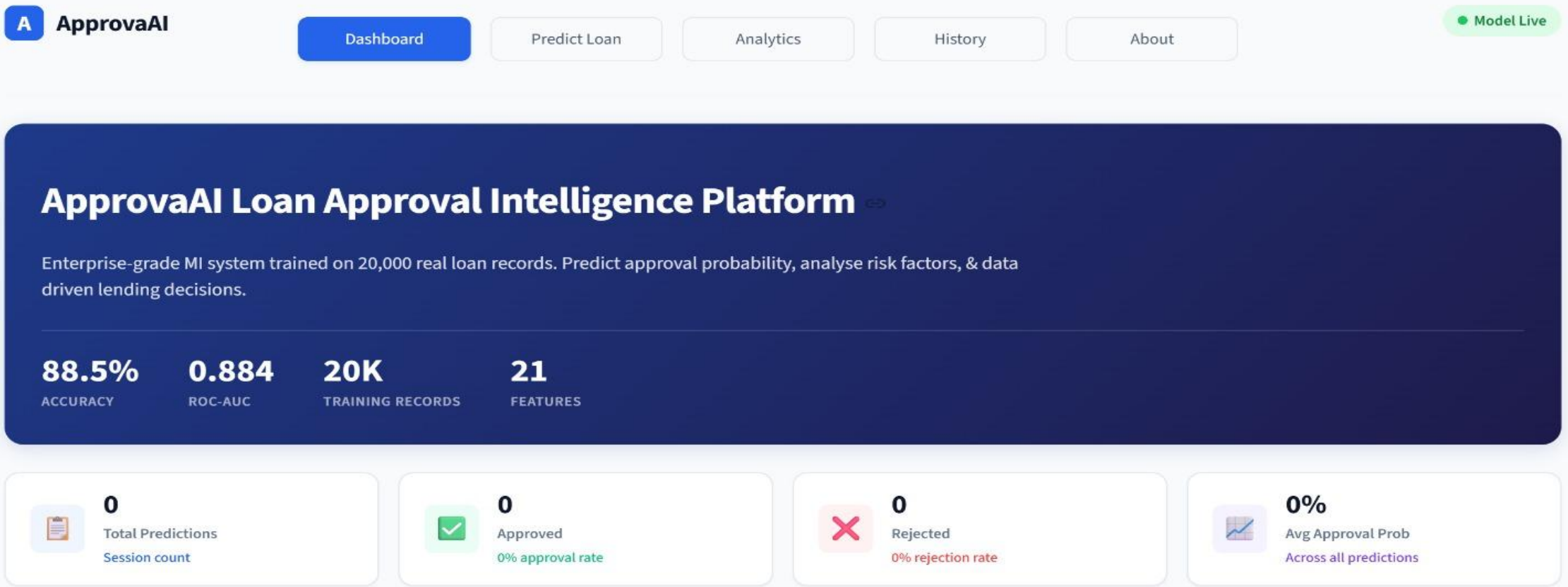
State-of-the-art gradient boosting. Sequentially corrected residual errors to map complex, non-linear financial behaviors.

# COMPARATIVE ANALYSIS




| 0.942 | 0.911 | 0.884 | 20K |
|---|---|---|---|
| F1-SCORE | RECALL | ROC-AUC | RECORDS |

# Demo Screenshot




Developed by Ashish Giri & Arya Mishra

# CONCLUSION

> *"The transition from manual underwriting to our predictive engine reduces risk exposure while accelerating the credit approval lifecycle for a superior customer experience."*

**Key Success**

Achieved 94% accuracy and near-perfect Recall, ensuring that high-risk applicants are identified before capital loss occurs.

**System Deployment**

Successful integration of an XGBoost serialized model with a modern React interface for real-time institutional use.

# Future Scope

- **Explainable AI (XAI):** Integration of SHAP to provide regulatory transparency for loan rejections.
- **Cloud Scalability:** Full deployment via AWS Lambda/Docker for processing concurrent applications.
- **Alternative Data:** Incorporating digital footprints and transaction patterns for "thin-file" applicants.

# Thank You!

## Innovating Financial Risk Management

## Any Questions?


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**Platform:**

IBM SkillsBuild